

Winning the Space Race with Data Science

Predicting Falcon 9 First-Stage Landing Success

Data collection • Wrangling • EDA • SQL • Folium • Dash • Classification

Executive Summary

Objective, methods and key results

Business objective: estimate whether a Falcon 9 first stage will land successfully, helping a competing launch provider assess launch risk and bid strategy.

Collected and consolidated 90 Falcon 9 launch records covering 2010–2020 using the course-era SpaceX dataset; demonstrated REST API and web-scraping workflows.

Performed data wrangling, visualization and SQL EDA; added geographic analysis with Folium and an interactive Plotly Dash design.

Compared Logistic Regression, KNN, SVM and Decision Tree classifiers. On a fixed 20% holdout, all four achieved 0.778 accuracy in this reference implementation.

Key insight: landing outcomes improve dramatically over time, with booster reuse, mission profile and launch-site context strongly associated with success.

Introduction

Business context and problem statement

SpaceX reduced launch economics through reusable Falcon 9 first stages. A competitor can use historical launch outcomes to understand the probability of successful recovery.

Problem statement: build a binary classifier where Class = 1 means successful first-stage landing and Class = 0 means unsuccessful/no landing.

Business metric: classification accuracy, supported by confusion-matrix analysis to distinguish false positives from false negatives.

Analysis horizon in this reference dataset: June 2010 through November 2020.

Data Collection — SpaceX REST API

Course-aligned extraction workflow

Endpoint used in the course workflow: <https://api.spacexdata.com/v4/launches/past>

Workflow: GET response → JSON normalization → select Falcon 9 records → extract launch, payload, booster and landing fields → create analysis dataframe.

Key fields retained: FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude and Class.

Repository URL required by the grader: upload this project folder to your GitHub account and replace the placeholder in README/presentation with your actual URL.

Data Collection — Web Scraping

Historical launch records with BeautifulSoup

Course workflow: request the historical Falcon 9 launch table → parse HTML with BeautifulSoup → identify table headers → extract row values → construct a Pandas dataframe.

This independent package includes a local HTML snapshot so the notebook executes deterministically without depending on a live Wikipedia page.

Parsed records are exported as `spacex_web_scraped.csv` and can be compared with the API-derived dataset.

Data Wrangling Methodology

Cleaning, transformation and feature preparation

Converted Date to datetime and standardized the analysis columns.

Checked missingness and replaced missing PayloadMass with the observed mean, following the course-era workflow.

Kept Class as the binary target: 1 = successful landing, 0 = unsuccessful/no landing.

Created a clean analysis table and a one-hot encoded modeling table for categorical variables such as Orbit and LaunchSite.

Exported clean_spacex.csv and dataset_part_3.csv for downstream EDA and machine learning.

EDA & Data Visualization Methodology

Questions answered with charts

How does landing success vary by launch site?

Does payload mass appear related to landing outcome?

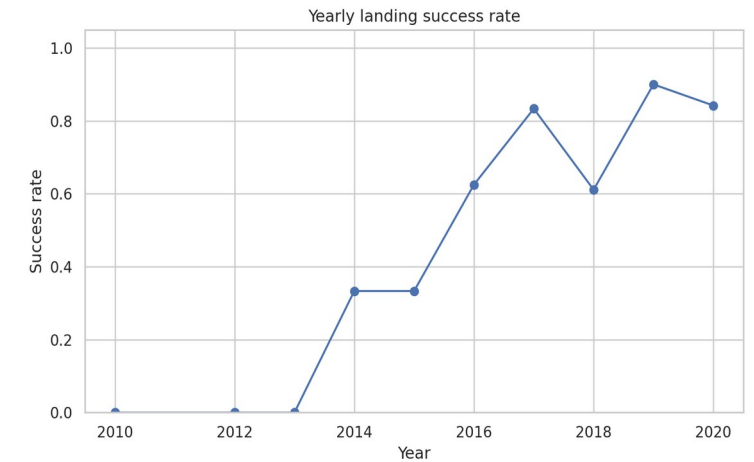
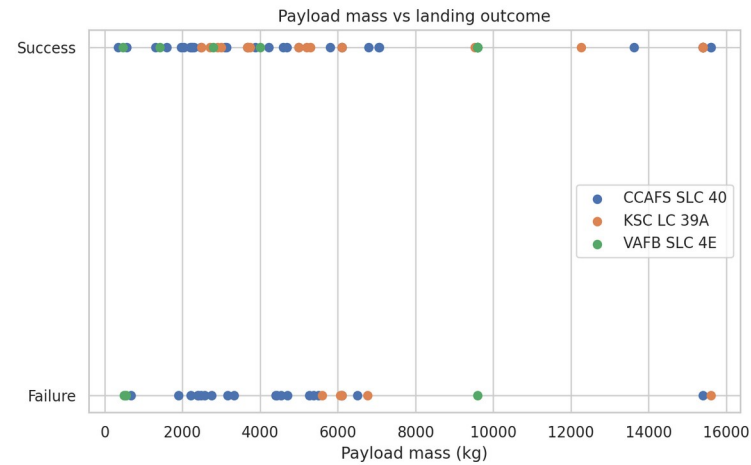
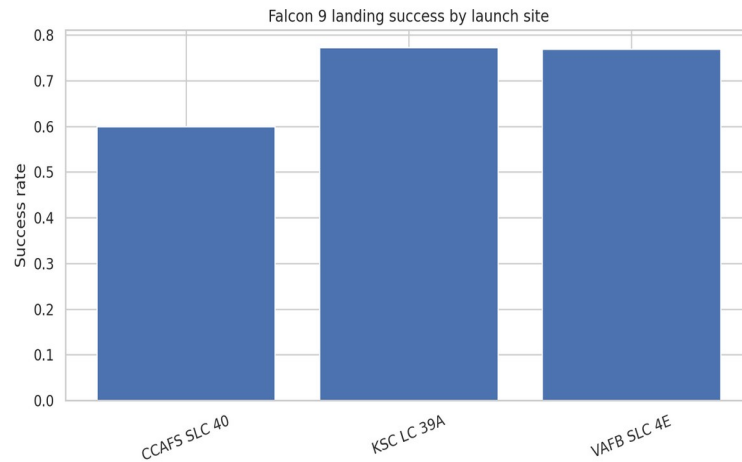
How has landing success changed by year?

Which orbital profiles have higher observed landing success?

Visualization stack: Pandas + Matplotlib/Seaborn for static analysis; Plotly for interactive views.

EDA Results — Visualization

Success rate, payload and time trends



Launch-site performance differs materially across the three U.S. sites in the dataset.

Payload mass is informative but not sufficient by itself to separate success from failure.

The strongest pattern is temporal: landing success rises sharply as reuse technology matures.

EDA with SQL — Methodology

SQLite analytical queries

Loaded the cleaned dataset into an in-memory SQLite database.

Queries covered: launch counts by site; average payload by site; success rates by site; yearly launch and success trends; maximum payload rankings.

SQL provides a compact, auditable way to reproduce stakeholder questions without relying solely on Pandas transformations.

Example: `SELECT LaunchSite, ROUND(100.0*AVG(Class),1) AS success_rate_pct FROM spacex GROUP BY LaunchSite;`

EDA with SQL — Results

Stakeholder-oriented findings

Launch volume: CCAFS SLC 40: 55, KSC LC 39A: 22, VAFB SLC 4E: 13.

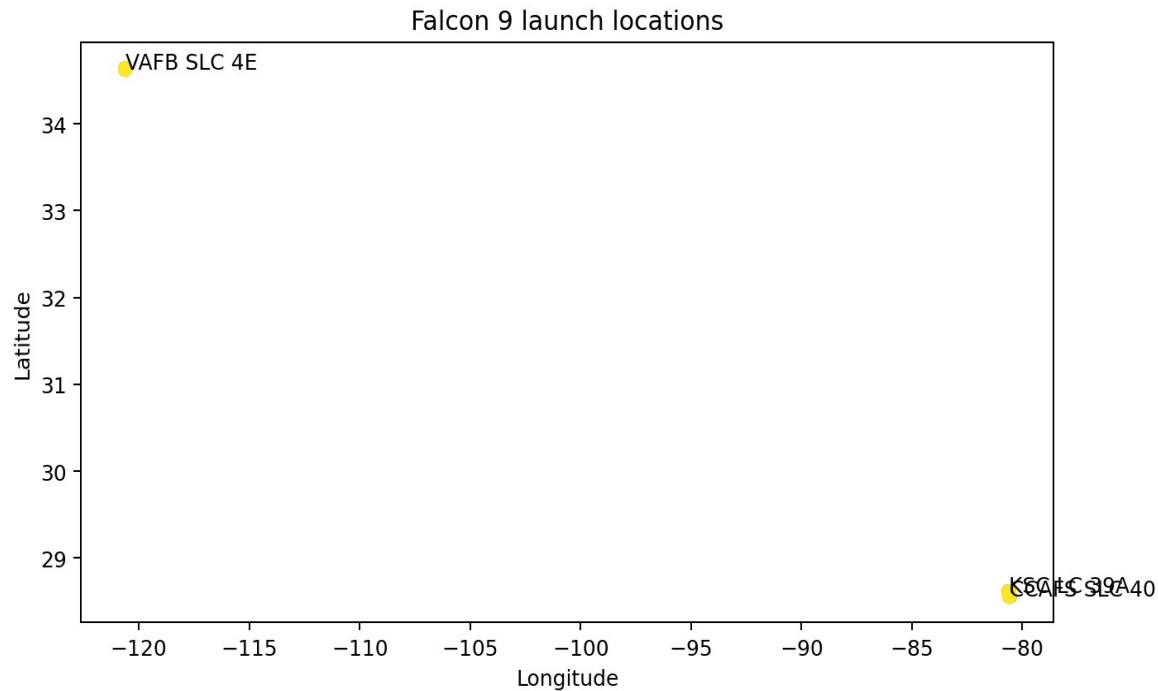
Success rates: KSC LC 39A: 77.3%, VAFB SLC 4E: 76.9%, CCAFS SLC 40: 60.0%.

Yearly analysis confirms the transition from mostly unsuccessful early recoveries to consistently high success rates in 2019–2020.

SQL result files are included in the data/ directory for reproducibility.

Interactive Visual Analytics — Folium

Launch-site geography and proximity analysis



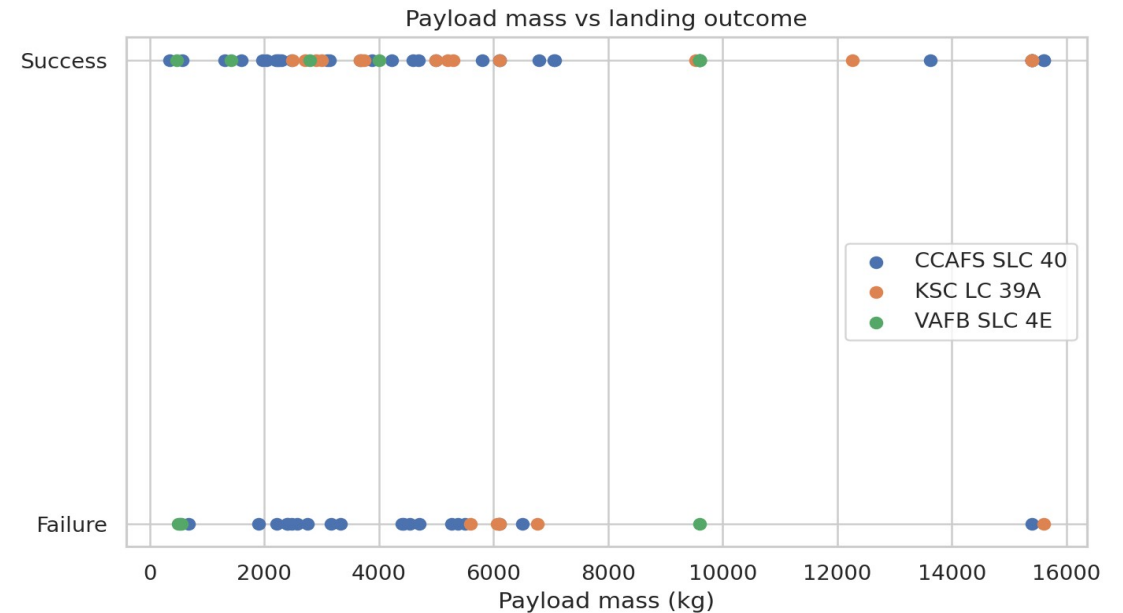
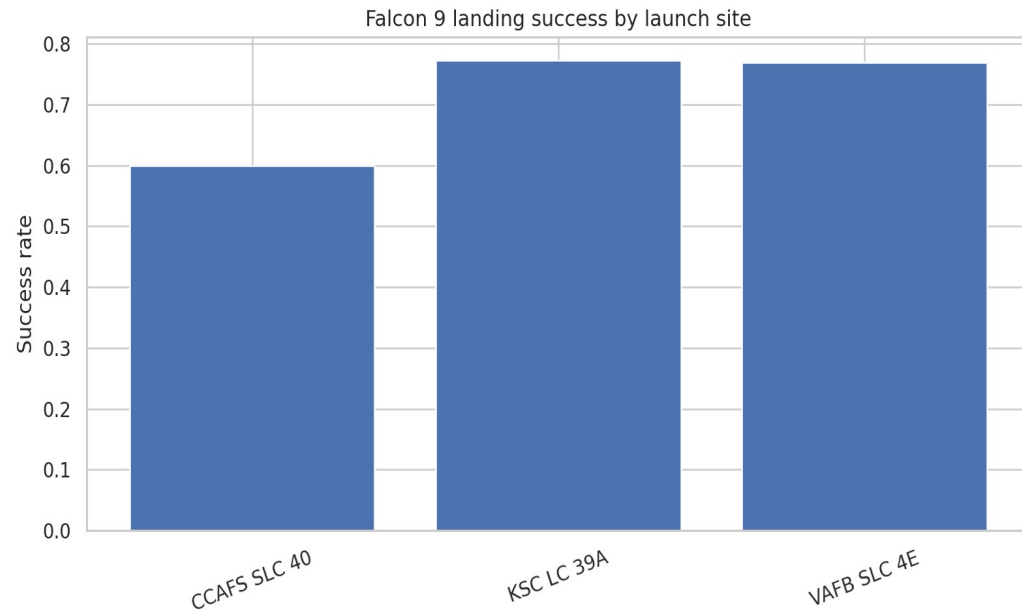
Three launch complexes appear in the dataset: CCAFS SLC 40, KSC LC 39A and VAFB SLC 4E.

Folium map includes launch markers, popups and a marker cluster for interactive exploration.

The actual interactive map is saved as `plots/spacex_launch_sites_map.html`.

Interactive Visual Analytics — Plotly Dash

Dashboard design and stakeholder use



Dashboard control: filter by launch site.

Primary KPI views: total launches and successful landings.

Visuals: success distribution and launch outcome by flight number / payload context.

Executable app.py is included and uses Dash + Plotly.

Predictive Analysis — Classification Methodology

Model development and evaluation

Feature set: payload mass, flight count, grid fins, booster reuse, landing legs, block, reuse count, orbit and launch site.

Categorical features were one-hot encoded; standardized pipelines were used for Logistic Regression, KNN and SVM.

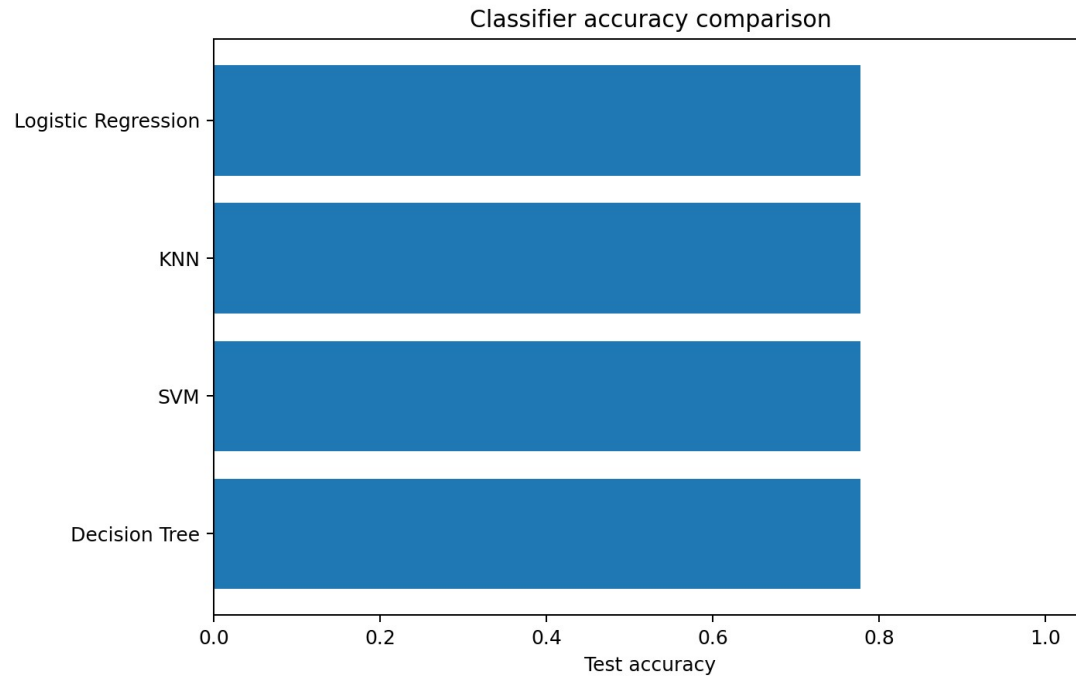
Compared four classifiers: Logistic Regression, KNN, SVM and Decision Tree.

Used an 80/20 stratified train/test split with `random_state=42` for reproducibility.

Evaluation: test accuracy, classification report and confusion matrix; permutation importance used for interpretation.

Predictive Analysis — Results

Model comparison and confusion matrix



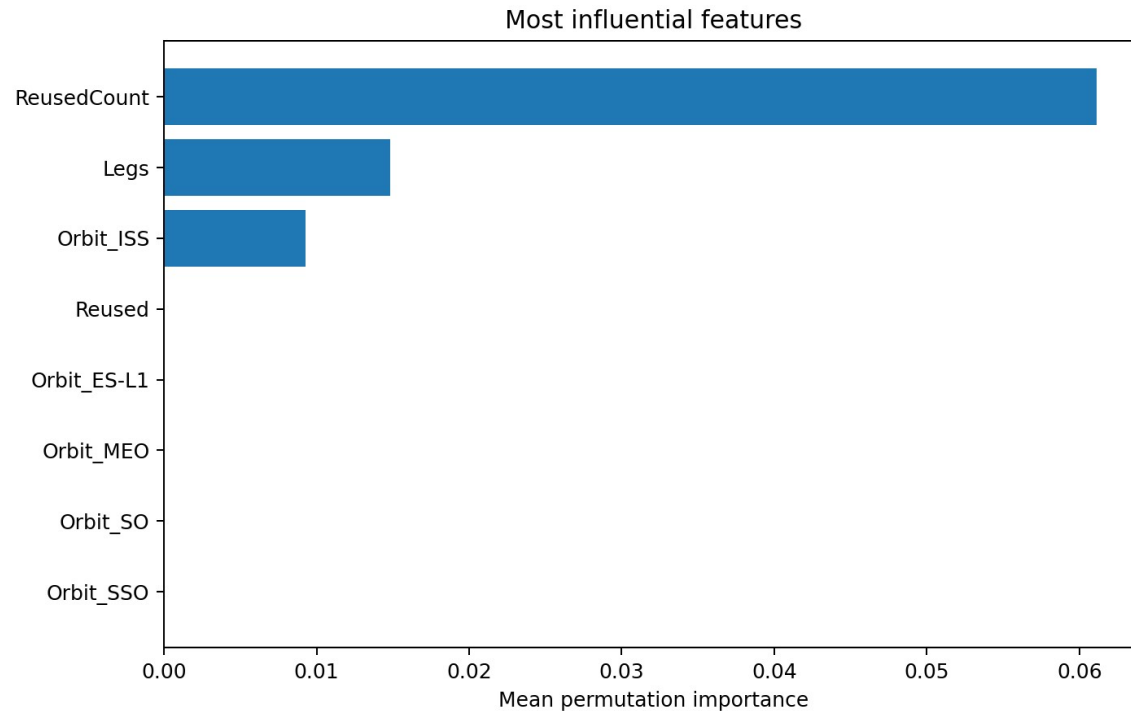
All four models reached 0.778 test accuracy on the fixed holdout.

The confusion matrix shows 12 successful landings correctly identified and 4 false positives; no successful landing was missed in this split.

With a small sample, the result should be treated as directional rather than as a production forecast.

Predictive Analysis — Best Model Explanation

What the model learns from the data



Permutation analysis highlights ReusedCount and landing-related configuration variables as useful signals.

This is consistent with the engineering story: repeated booster use is associated with increasing operational maturity.

The model should be refreshed with newer launches and evaluated with time-aware validation before real bidding decisions.

Conclusion

Business takeaway

The historical data contains strong evidence that Falcon 9 first-stage landing reliability improved over time.

Launch site, mission profile and booster history are useful predictors of landing outcome.

The classification models demonstrate a workable proof of concept, but the small dataset limits generalization.

For a competing provider, the analysis can support risk-adjusted launch pricing, site selection and scenario planning.

Next step: incorporate newer SpaceX API records, richer weather/mission variables and time-based validation.

Creative & Innovative Insights

Beyond the basic rubric

“Reliability curve”: treat landing success as an operational maturity signal rather than only a binary label.

“Reuse premium”: repeated booster flights are associated with stronger observed outcomes, suggesting reuse history can act as a proxy for system maturity.

“Site context matters”: launch-site geography and mission profile can be combined with model predictions to build risk-adjusted bid scenarios.

“Model honesty”: when several classifiers tie on the holdout, the correct decision is to report the tie rather than overstate a marginal winner.

Appendix — Deliverables & Repository

Files included in the reference package

8 executed Jupyter notebooks covering API, web scraping, wrangling, SQL, visualization, Folium, Dash and classification.

Data: dataset_part_1.csv, dataset_part_2.csv, dataset_part_3.csv, clean_spacex.csv and dashboard/map extracts.

Interactive: plots/spacex_launch_sites_map.html and app.py.

Presentation: this PDF plus the editable PPTX.

GitHub URL: PASTE YOUR OWN REPOSITORY URL HERE after uploading the project folder to GitHub.